Two questions: what score, and where the atoms go

One binary variable = one atom. Fewer variables means a smaller, more faithful register.

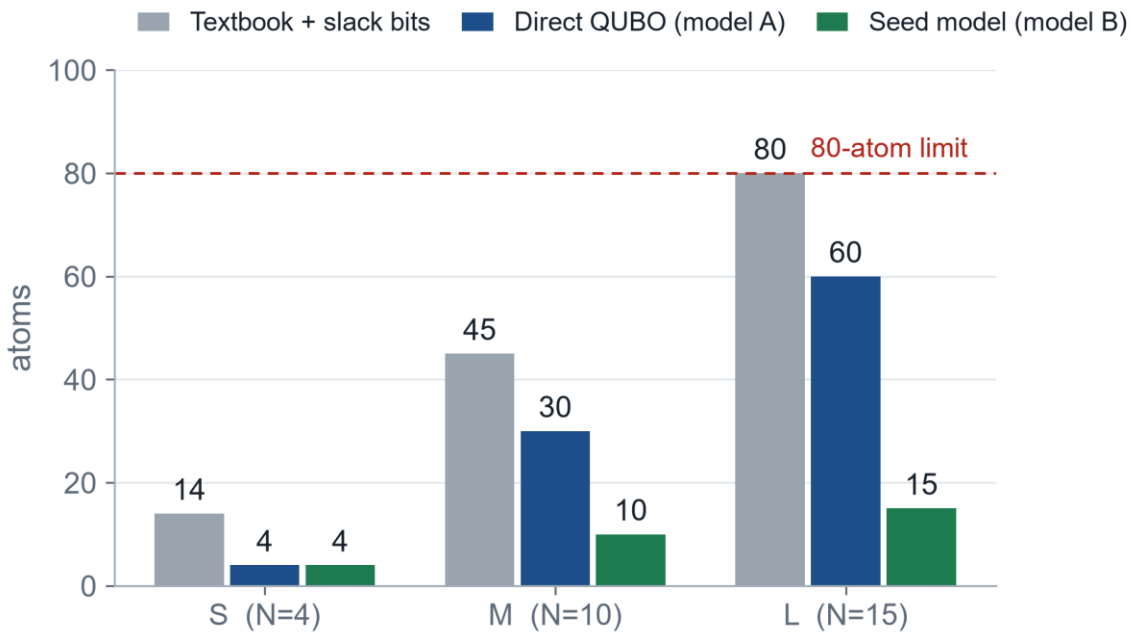
A1 · design the QUBO

Write one score for every bitstring.

Lowest score = best valid grouping.

A3 · lay out the atoms

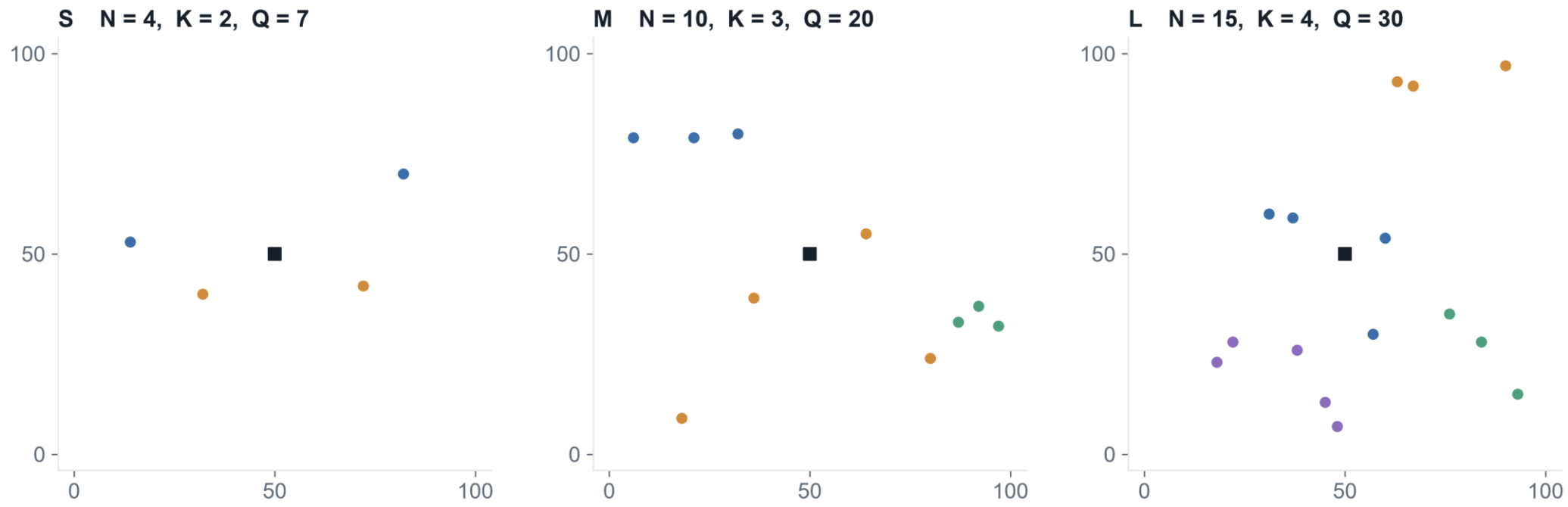
Place atoms so the machine’s energy gives the same scores, and report how far off it is.



A2 instances and **A4** pulse support these two.

Nine certified instances, each with a proven answer

3 seeds per tier. Every instance ships a feasible certificate and an exact optimum (DP, checked by CPLEX).



	S	M	L
customers N / trucks K / capacity Q	4 / 2 / 7	10 / 3 / 20	15 / 4 / 30
proven optimum C* (seed 0, cluster cost)	110.1	353.0	475.4

Colour = truck in the proven optimum; black square = depot.

Cluster is optimised; capacity only has to pass the line

Example: customers A, B, C, D with demands 3, 4, 2, 3; 2 trucks of capacity 7. Bit 1 = truck 1, bit 0 = truck 2.

① $$\text{Total Loss} = C_{\text{cluster}} + \lambda P_{\text{capacity}}$$

② $$C_{\text{cluster}} = \sum_{i<j} d_{ij}\,(1 - x_i - x_j + 2x_i x_j)$$

Pays d_{ij} only when i and j share a truck: the bracket is 1 for 00 and 11, 0 for 01 and 10.

③ $$P_{\text{capacity}} = \left(\sum_i q_i\, x_i - \frac{D}{2}\right)^2$$

q_i = demand, D = total demand. No slack bits: valid splits score 0–1, an overloaded truck scores ≥ 4, and λ makes that too expensive.

③ capacity on the example

bits ABCD	loads	P	valid?
1001	6 | 6	0	**yes**
1100	7 | 5	1	**yes**
1010	5 | 7	1	**yes**
1110	9 | 3	9	**no**
1111	12 | 0	36	**no**

Honest cost: among valid splits it still prefers 6 | 6. When the best split is unbalanced this loses objective (M_seed2: 13.8 %, before any quantum step).

Same two slots, but not the same freedom

Expanded, our loss is one number per atom plus one per atom pair. Pasqal's energy has exactly those two slots.

④ $$\text{Total Loss} = \text{const} + \sum_i a_i\, x_i + \sum_{i<j} b_{ij}\, x_i x_j$$

$$a_i = -S_i + \lambda\, q_i (q_i - D) \qquad b_{ij} = 2d_{ij} + 2\lambda\, q_i q_j$$

S_i = sum of customer i's distances to everyone else.

⑤ $$E = \sum_i (-\delta)\, x_i + \sum_{i<j} \frac{C_6}{r_{ij}^6}\, x_i x_j$$

δ = laser detuning (one value for all atoms); r = distance between atoms.

⑥ term	our QUBO	Pasqal	match?
per atom	**a_i, different per customer**	−δ, the same for all	**no**
per pair: value	**b_{ij}, many different values**	C_6 / r^6, set by distance	**approx.**
per pair: sign	positive	positive	**yes**

On M and L the per-atom mismatch is harmless on valid answers (each customer has exactly one bit on, so Σ $a_i x_i$ is a constant).

The real problem is the pair slot: **some pairs want 0, but $1/r^6$ is never 0.**

Atom distance: S already distorts, M cannot be laid out

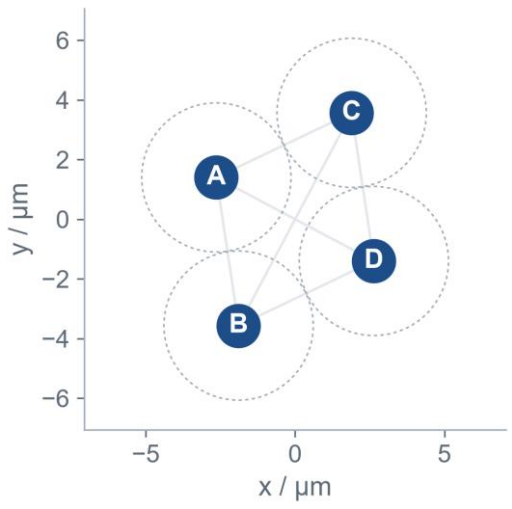
Each wanted coupling gives a wanted distance. The question is whether all of them fit in one 2D layout at once.

⑧ $$r = \left(\frac{C_6}{\text{wanted coupling}} \right)^{1/6}$$

r ≥ 5 µm

radius ≤ 38 µm

⑦ $$\text{error} = \text{Pasqal value} - \text{our value (scaled)}$$

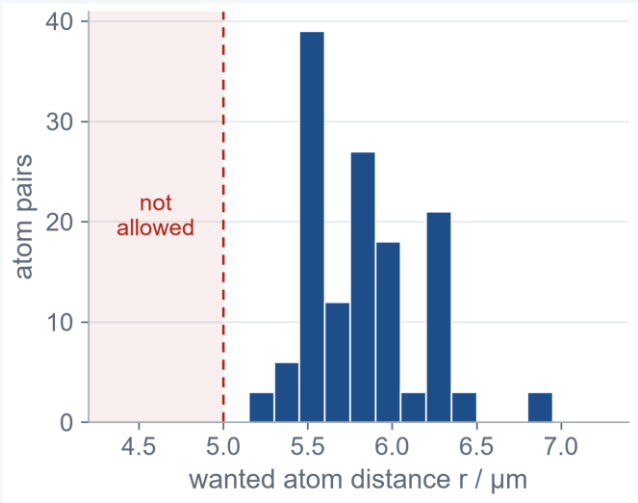


S · 4 atoms

Legal layout found.

Coupling error η = 0.54

But the true optimum already ranks #8 of 16 under the realised energy.



M · 30 atoms

Every pair wants 5 to 7 µm. Each atom wants 11 partners that close.

A plane at 5 µm spacing fits about 6 neighbours that close.

Measured: the true optimum ranks ≈ 10^4 of 59 049; η ≈ 1.

Change the question: the atom map is the customer map

Brief §4.1: “avoid the inequality altogether by changing what the model asks for.”

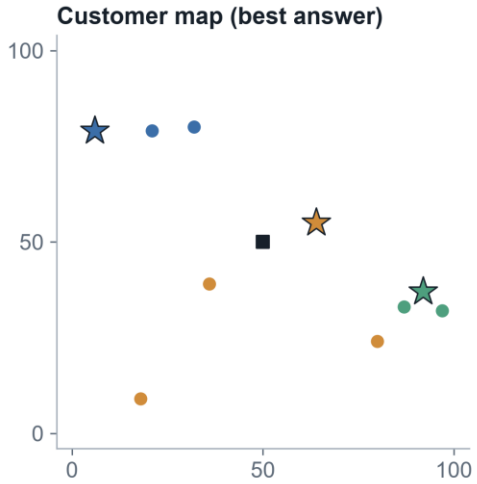
$x_i = 1 \Leftrightarrow$ customer i anchors a truck. N atoms: 4 / 10 / 15.

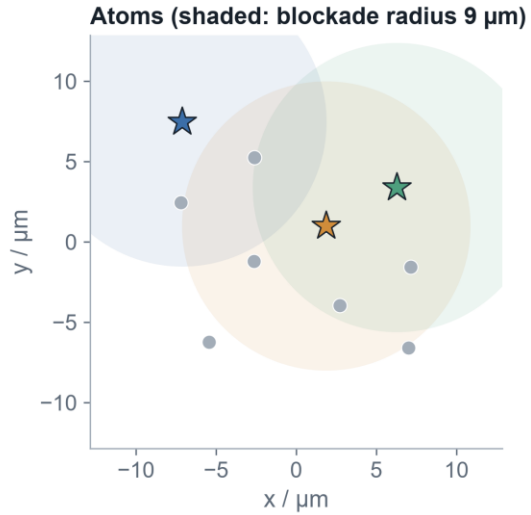
$$E = -\delta \sum_i x_i + \sum_{i<j} \frac{C_6}{(\alpha\, d_{ij})^6} x_i x_j$$

term	model B wants	Pasqal gives	match?
per atom	the same −δ	−δ, the same for all	**yes**
per pair	$C_6 / (\alpha\, d_{ij})^6$	C_6 / r^6 at $r = \alpha\, d_{ij}$	**yes**
capacity	—	not in the energy	**decode**

Close customers block each other, so the lowest energy picks K spread-out seeds.

Then every other customer joins the nearest group with room (light decode, classical, O(N²K)).

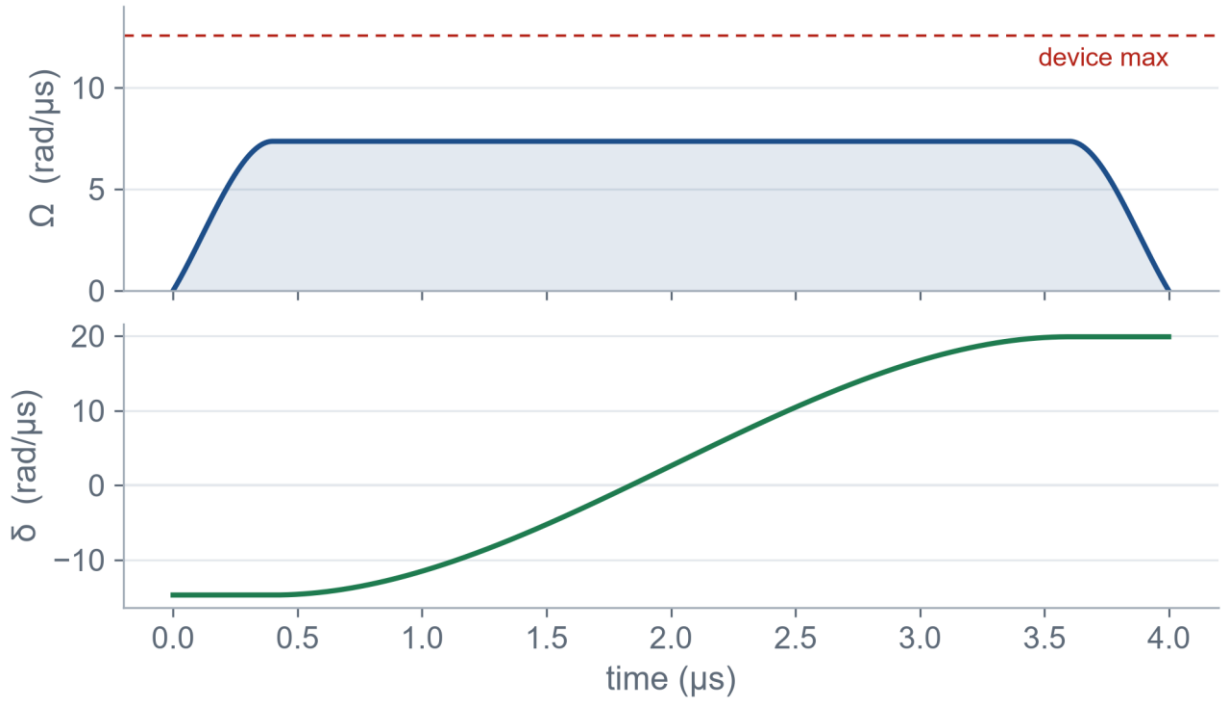




M_seed0: η = 0.72; 12 of 120 seed sets decode to the optimum.

Pulse: Ω and duration per tier, final δ per instance

Standard adiabatic sweep. Every sequence is built and checked by Pulser against AnalogDevice.



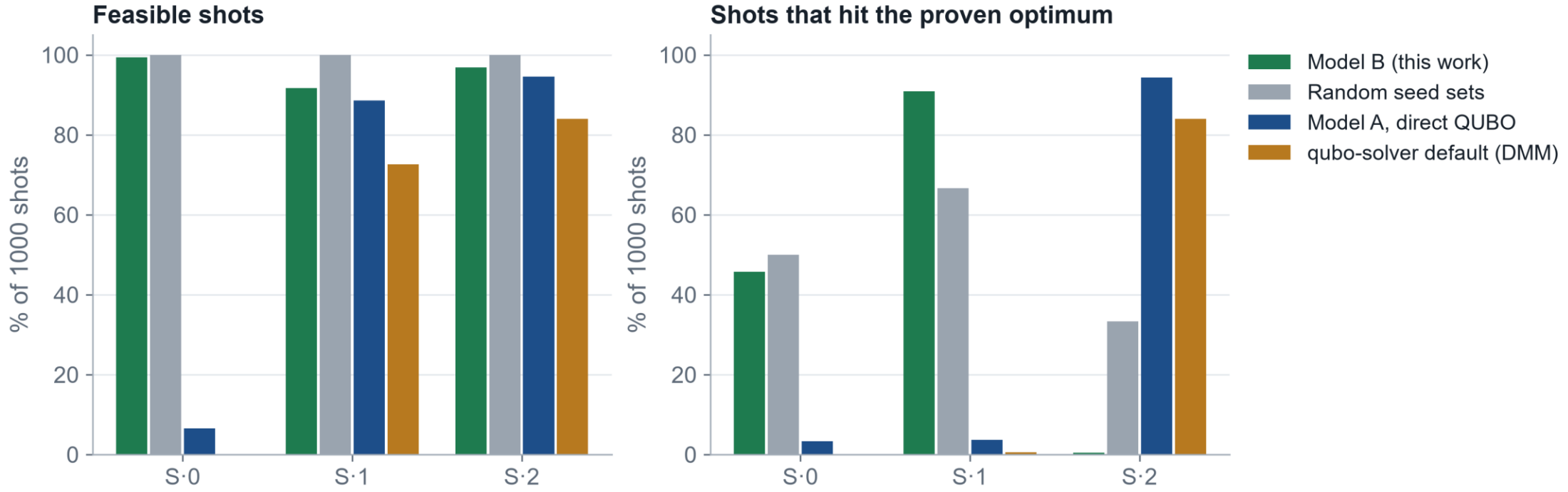
S_seed0 as run: Ωmax = 7.36 rad/µs, T = 4000 ns, δ: -14.7 → 19.9 rad/µs.

parameter	what it does	how we set it
final δ	the per-atom term	midpoint of the window where exactly K seeds is lowest
Ω (strength)	lets atoms flip; sets the blockade radius	grid per tier: radius ∈ {7, 9, 12} µm
duration T	longer = closer to the best answer	grid per tier: {2000, 4000, 6000} ns
δ scale c	fine-tunes the final δ	grid per instance: c ∈ {0.7, 1.0, 1.3}
shots	more = more reliable rates	1000

Selection score: share of shots that hit the optimum, then best gap, then feasibility (brief §8.2: reward good shots, not the mean).

Tier S is too small to beat chance; M and L decide

S has only 6 seed sets, so random picks already hit the optimum 33–67 % of the time. 1000 shots, global detuning unless marked.



Did not work

Direct QUBO on M/L: no 2D layout reproduces it (η ≈ 1). Load-balance penalty loses up to 13.8 % before any quantum step.

Caveats

Light decode is order-sensitive (S·2: 0.4 % vs 96 % with exact decode). AI assisted code, figures and drafting; every number is from results/.